

Field Task & Production Management

Phase 0 — Action plan, milestones and build package · **iOS edition**

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Client Native iOS (SwiftUI) **Backend** Supabase Postgres **Issued** 20 September 2026

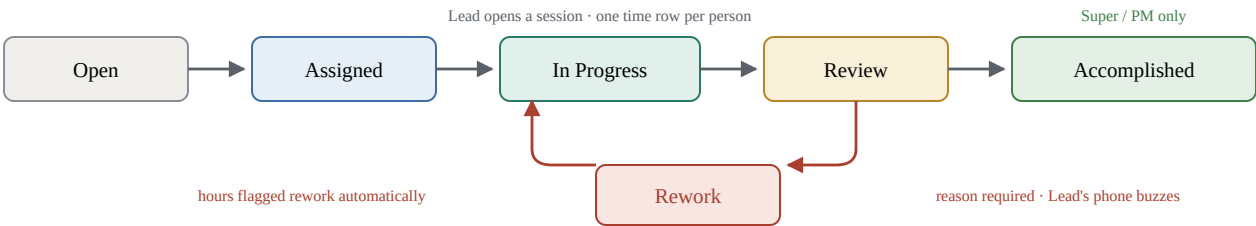
1. What changed, and what did not

The original package specified a React mobile-first PWA. The client is now a **native iOS app**. Everything underneath is unchanged — the schema, the row-level security, and all nine non-negotiable principles never depended on the client.

REQUIREMENT FROM THE PHASE 0 RISK LIST	PWA ON IOS	NATIVE
Rework notification must reach the Lead with the app closed	Fragile — needs Home Screen install first	APNs. Reliable.
Close-out under 60 seconds, camera one tap	File-picker detour	Direct camera
A close-out with no signal must not be lost	Browser storage evicted under pressure	SwiftData, survives
Crews shoot 12 MP images all day	JavaScript-side compression	Hardware pipeline

The cost, stated plainly. No Android. App Store review on every release. A Mac and an Apple Developer account (\$99/yr) are now required to ship. Distribution to the Leads during the pilot is TestFlight, not the public store — which is faster, but caps you at 100 internal testers and expires each build after 90 days.

2. Status flow — unchanged, and locked in the database



Six core statuses, enforced by a Postgres enum. A seventh cannot be inserted from any client. Custom board columns sit on top and each maps to exactly one of these, so reporting reads core status and never a column name.

3. Phase plan and milestones

#	DELIVERABLE	MILESTONE — CHECKED WHEN	STATE
0	Plan, model, flow, checklist	This package issued	DONE

1	Schema · auth · admin user setup · row-level security	Admin creates a Lead by email; that Lead sees only his project and cannot read the receipts tables at all	DATABASE GATE MET 56/56 assertions green. iOS shell awaiting a Mac build.
2	Task board, columns, role-gated transitions	A Lead cannot move a task to Accomplished	NEXT
3	Time capture — crew selection, per-person rows	One session with 3 crew writes 3 rows	—
4	Photos, description, quantity at close-out	Task closes with a photo from a phone in the field	—
5	Review gate · Rework column · push	Rejected task lands in Rework and the Lead's phone buzzes	—
6a	Cost class, change orders, custom columns	CO hours excluded from base earned value	—
6b	Schedule fields, GC baselines, blocked periods	Blocked days attribute to a named party	—
6c	Work class, discovered defects, source links	A defect ticket links back to an Accomplished task	—
6d	Material page — upload, extraction, categories	A photographed receipt lands categorised, duplicates caught	—
7	Progress page, every number drillable	Clicking any figure returns its ticket list; numbers reconcile to a hand-check of one week	—
8	Pilot on a fresh PCM job	Two full weeks run without paper backup	—
9	Closeout — results, improvements, business case	Final PDF issued	—

4. Phase 1 — what is built and proven

The Phase 1 milestone is a database claim, and it is now tested rather than asserted. Fifty-six assertions run as the role a phone actually holds — `authenticated` , with a real user id — not as the database owner, who bypasses every policy. The suite exits non-zero when a rule regresses, so it gates CI.

The matrix, proven

AREA	TESTS	WHAT IS ACTUALLY ASSERTED
Project isolation	9	A Lead sees exactly one project. A sub's Lead on another job sees none of the first job's tasks. A Manager sees every project. The same man is Foreman on one job and Super on another.
Material access	15	Lead reads zero receipts, zero line items, zero categories. A Foreman reads his own receipt and cannot reach the other Foreman's by id. PM, Assistant PM and Manager read everything.
Sign-off	4	Lead, Foreman and Assistant PM are all refused. Super is allowed.
History	7	An Accomplished task cannot be reopened, by anyone, including the Manager. <code>task_events</code> cannot be rewritten even by the service role.
Change orders	4	Assistant PM and Super refused in both directions. PM allowed.
Locked statuses	5	A seventh status is impossible. A column cannot map to a status it does not match. <code>rework_count</code> cannot be typed, only derived.
Lead movement	4	A Lead moves his own task and cannot touch another Lead's, create a task, or kick work back.
People & cost	8	A Lead cannot read <code>burdened_rate</code> at all. Nobody can be deleted — deactivation is the only path.

Six defects found and fixed in the reviewed schema

Four of them would have stopped the app working entirely. Full write-up in `docs/DECISIONS.md` .

REF	DEFECT	EFFECT IF SHIPPED
FIX-1	<code>citext</code> created after the table that uses it	Migration could not apply at all
FIX-3	<code>auth.uid()</code> is null for service-role writes, but <code>actor_id</code> is NOT NULL	Every seed and background status change fails
FIX-4	The rework counter trips the guard that protects its own column	No kick-back can ever be recorded
FIX-5	Cost view unusable as <code>security_invoker</code> under column grants	Nobody sees burdened rates, PM included

FIX-6	The audit trigger is blocked by its own table's RLS	No status change saves, for any role, with no error that points at the cause
FIX-2	<code>qty_needs_unit</code> ignored <code>qty_installed</code>	Close-out records "340" with no unit

5. Reminders & checklist

Before any more code — these block the build

- ☐ **Name the app and lock it.** Blocks the bundle identifier, the App Store listing and the repo name. Currently `com.apexmech.fieldtask` as a placeholder.
- ☐ **Apple Developer Program enrolment** (\$99/yr). Takes up to 48 hours; start it before it is on the critical path.
- ☐ **A Mac with Xcode 15 or later.** Nothing iOS ships without one.
- ☐ **Create the Supabase project** (free tier is sufficient for the pilot) and apply `0001` , `0002` , then seed.
- ☐ **Confirm which iPhones the Leads actually carry.** iOS 17 is assumed; an iPhone 8 or SE 1 in the crew drops the target to 16.

Before the pilot goes live

- ☐ Written OK from PCM ownership before any PCM data lives in an Apex-owned system.
- ☐ Decide data ownership up front: who owns the records if the relationship ends.
- ☐ Decide burdened rates: real numbers, or placeholder multipliers until PCM shares them.
- ☐ Confirm the area / system list matches how PCM actually breaks out its estimates.
- ☐ Confirm the material category list against PCM's estimate breakdown.
- ☐ Agree the root-cause list for defects — short and fixed, never free text.
- ☐ Get the GC's published schedule in a usable form, or accept manual milestone entry per area.
- ☐ TestFlight build in every Lead's hands, with notifications actually tested on their phone.
- ☐ 15-minute walkthrough with each Lead — one screen, one rule: close the task before you leave.
- ☐ Run one week in parallel with existing daily reports and compare totals.
- ☐ CSV export path confirmed out of every table, so nothing is trapped.

Standing rules during the build

- ☒ Core statuses locked in the database, not hidden in the UI.
- ☒ Every status change writes to `task_events` , no exceptions.
- ☒ Never allow direct edit of hours; only start/end times and crew selection.
- ☒ Row-level security enforced in the database, proven by test, not by the UI.
- ☒ Metrics built as saved filters from the start, so drill-down is free rather than a rewrite.
- ☐ Phone-width layout first, one-handed, portrait.
- ☐ Offline tolerance: a close-out with no signal must not be lost.

- ☐ Photo compression on upload — receipts must stay legible enough to audit against.
- ☐ Rework notification must reach the Lead with the app closed.

6. Risks still live

RISK	HANDLING
Leads don't close tasks in the field	Close-out under 60 seconds: crew already selected, camera one tap, quantity one field. Measured on a real phone in Phase 4, not assumed.
PCM says "that's just part of your job"	Built and owned under Apex, on Apex infrastructure, on your own time. PCM is the pilot customer, not the owner. In writing before go-live.
Rework becomes a blame tool	Reasons describe the condition, not the person. Defect rate is never shown per individual — by system, area and phase only. This is built in, not policy.
Scope creep into a full ERP	v1 ships with no scheduling, no invoicing, no payroll export, no Android.
App Store review delays a fix	TestFlight for the pilot; expedited review exists for genuine blockers. Budget 24–48 hours for any store release.
Data trust	Phase 7 is not done until a week of app numbers matches a hand tally.

Next artifact: Phase 1 completion PDF — full chat interactions plus summary — issued once the iOS shell has been built on a Mac and a real Lead has signed in against a live Supabase project. The database half of Phase 1 is already proven and is recorded above.